

# TRAVELING SALESPERSON PROBLEM

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Monte Carlo Methods  
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24. Oktober 2024

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# 1 Introduction

The traveling salesman problem is a popular entry problem for computational optimization using probabilistic methods. It was first formulated in 1930 and is since then used as a benchmark for many optimization algorithms. Even though the problem is computationally difficult, many heuristics and exact algorithms are known, which can approximate the exact solution within a small fraction of 1 %. This document focuses on implementing two different approaches to finding the optimal path and then comparing their runtime and uncertainties their calculations. First we focus on implementing a simple Metropolis algorithm with simulated annealing, where we compare different annealing strategies with the use of temperature functions. The second part of this document a genetic algorithm is implemented and at last compared to the probabilistic approach.

## 2 fundamentals

### Literatur